Define Problem Statements

Phase 1: Brainstorming & Ideation

1. Background

A trip plan combines many decisions: where to go, when, how much to spend, who is travelling, how fast to travel and what to eat. A traveller may have all the information and still struggle to turn it into a realistic sequence of days, especially when the group has different interests or a fixed budget.

2. Primary Problem Statement

Travellers need a simple way to turn their destination, dates, group size, budget, travel style, pace, interests and dietary needs into a structured day-wise itinerary with a running cost estimate, which they can edit, save and print, without building every day of the plan by hand from scattered sources.

3. Related Problems

Scattered planning inputs

Preferences are often kept in notes and chats. A single guided wizard collects them in one place and gives the AI service a consistent input.

Lists instead of schedules

A list of attractions does not say how to spread activities across days. The itinerary groups activities into morning, afternoon and evening slots with start times and durations.

Budget surprises

Costs are usually discovered late. The budget summary compares the target budget with the estimated total (lodging, food, activities, transit) and shows a warning when the plan is over budget.

Plans that cannot be adjusted

A generated plan is rarely perfect. The edit page allows activities to be changed or removed, and a day can be regenerated; the budget is recalculated from the edited plan.

Returning to an earlier plan

The Saved Trips page lists stored trips with search and a travel-style filter so that a plan can be reopened without generating it again.

4. Objectives

1. Collect trip preferences through a five-step wizard with validation at each step.
2. Generate a structured itinerary using Gemini or OpenAI, with a heuristic fallback when no provider key is set.
3. Validate requests with Pydantic schemas and normalize the model response before it is shown.
4. Show a live budget breakdown and warn when the estimate exceeds the target budget.
5. Allow trips to be saved, searched, filtered, edited, regenerated by day, and deleted.
6. Support sign-in through Supabase Auth, with row-level security in the database, plus a one-click demo mode.
7. Allow the itinerary to be printed or saved as a PDF through the browser.

5. Expected Outcomes and Validation

The project should be judged by the behaviour of the whole workflow rather than by individual screens. A valid request should return a structured plan; the user should be able to review it, adjust it, save it, find it again and print it. Formal performance results, interviews and usability scores were not part of the supplied materials; they should be added only if actually collected. Generated costs are estimates, not verified prices.